

Electron vs. Hole Energy Dispersion

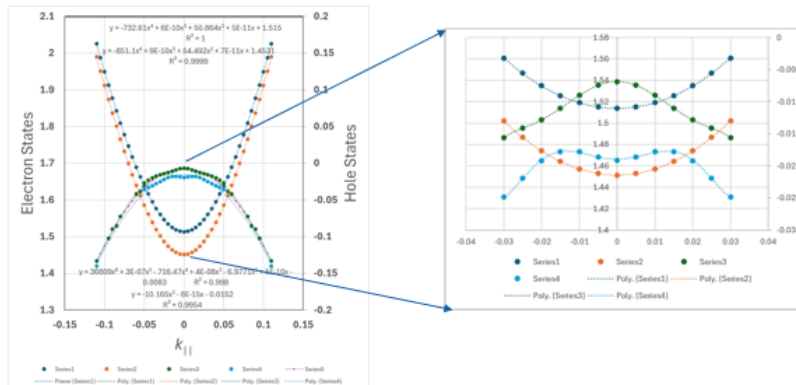
Why the parabolic effective-mass approximation works for electrons but not for the hole states on your boss's slide

One-sentence takeaway: Main conclusion: the electron subbands are close enough to parabolic that a constant effective mass is a useful approximation. The displayed hole subbands are not well represented by one parabola, so their full k-dependent energy relation must be modeled more carefully before integrating χ^2 .

Energy relations for electron and hole state

Free electron assumption $E_{b,m}(k_{||}) = \frac{\hbar^2 k_{||}^2}{2m_{b,m}^*} + E_{b,m}(k_{||=0})$ fitting energies with parabolic term allows m^* to be determined

Optel $k \cdot p$ Calculations



Conclusions

1. Electron State are more-or-less parabolic. Effective mass approximation should work!
2. Holes state energies are not parabolic in the free dimension. $E(k_{||})$ needs functional relation other than parabolic for integration.

Original slide supplied by the user.

1. What question is your boss testing?

The previous slide proposed a simple model for in-plane motion:

$$E_{b,m}(k_{\parallel}) = E_{b,m}(0) + \frac{\hbar^2 k_{\parallel}^2}{2m_{b,m}^*}$$

Because this equation contains $k_{\parallel}$ squared, it predicts a parabola in an energy-versus- k plot. Your boss is now comparing that simple model against a more complete $k \cdot p$ calculation to see whether it is actually accurate for the relevant electron and hole subbands.

Practical question: The test is not "are the bands exactly free-electron bands?" The practical test is "is a constant-effective-mass parabola accurate enough over the k range that contributes to the nonlinear-optical integral?"

The simple answer

State type	What the plot says	Consequence
Electron states	More-or-less parabolic	A constant effective mass approximation should be usable.
Hole states	Noticeably nonparabolic	Use a better $E_h(k_{\parallel})$ relation instead of forcing one constant mass over the full range.

2. What the free-electron / effective-mass model means

$$E_{b,m}(k_{\parallel}) = E_{b,m}(0) + \frac{\hbar^2 k_{\parallel}^2}{2m_{b,m}^*}$$

Read this as two energy contributions: the QW confinement energy at $k_{\parallel} = 0$, plus the kinetic energy associated with motion in the plane of the quantum well.

Term	Meaning
$E_{b,m}(0)$	The confined-subband energy at the zone center.
$\hbar^2 k_{\parallel}^2 / 2m^*$	The extra in-plane kinetic energy.
$m_{b,m}^*$	Effective mass for that particular band/subband.
$k_{\parallel}$	Magnitude of the in-plane wavevector.

Why fitting a parabola gives the effective mass

If the computed dispersion can be fitted as $E(k) = E(0) + A k^2$, then the coefficient A is directly related to the effective mass:

$$m^* = \frac{\hbar^2}{2A} \quad \text{when} \quad E(k) = E(0) + A k^2$$

$$\frac{d^2 E}{dk^2} = \frac{\hbar^2}{m^*}$$

Curvature rule: Strong curvature means a smaller effective mass. A flatter parabola means a larger effective mass.

3. How to read the plots on the slide

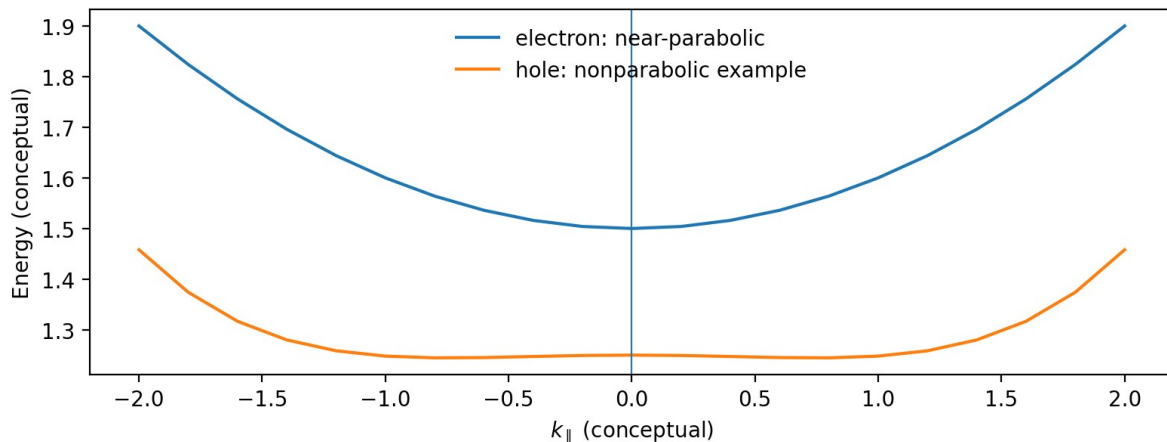
The colored points are calculated **k.p energies**. The fitted curves test whether a simple parabolic model can follow those points.

Electron states

- **Visual behavior:** the electron branches form smooth U-shaped curves around $k_{\text{parallel}} = 0$.
- **Mathematical implication:** $E_e(k)$ can be approximated by $E_e(0) + A_e k^2$ over the plotted range.
- **Physical implication:** one effective-mass parameter can reproduce most of the electron dispersion that matters for the integration.

Hole states

- **Visual behavior:** the hole branches change curvature and do not follow one clean parabola over the same range.
- **Mathematical implication:** a single A_h , and therefore a single constant m_h^* , cannot reproduce the full hole dispersion.
- **Practical implication:** the hole energies should be represented by the calculated dispersion itself or by a more flexible fit/interpolation.



Conceptual illustration only: the exact hole curve must come from the actual calculation.

4. Why the hole nonparabolicity matters for Equation 2

The energy in the nonlinear susceptibility denominator is not an isolated electron energy or hole energy. It is an interband transition energy:

$$\Delta E_{nm}(k_{\parallel}) = E_{e,n}(k_{\parallel}) - E_{hh,m}(k_{\parallel})$$

Therefore an error in the hole dispersion directly changes the transition energy that appears in the resonance denominators.

Why accuracy matters: Near resonance, even a modest energy error can produce a large change in $\chi^{(2)}$, because the energy difference appears in the denominator rather than as a small additive correction.

Example logic

1. Use $E_{e1}(k)$ and $E_{hh1}(k)$ to construct $\Delta E_{11}(k)$.
2. Compare $\Delta E_{11}(k)$ with the photon-energy combinations in the Eq. 2 denominators.
3. If an inaccurate hole parabola shifts $\Delta E_{11}(k)$, the calculated detuning changes.
4. The k values that appear resonant, near-resonant, or off-resonant can then be misweighted in the final integral.

5. What changes in the practical calculation?

For electrons

If the parabolic fit is validated, the electron dispersion can be compressed into two parameters for each subband:

- **Zone-center energy:** $E_{e,n}(0)$
- **Effective mass:** $m_{e,n}^*$

Then reconstruct:

$$E_{b,m}(k_{\parallel}) = E_{b,m}(0) + \frac{\hbar^2 k_{\parallel}^2}{2m_{b,m}^*}$$

Simplification: This is computationally convenient: a fitted effective mass replaces a large table of electron $E(k)$ values.

For holes

The slide says this simplification is not adequate. Instead, the calculation needs a more faithful relation $E_{hh,m}(k_{\parallel})$. The slide does not prescribe the exact form; it only establishes that one constant-mass parabola is not sufficient.

- Possible numerical representation: directly interpolate the computed k.p hole energies.
- Possible analytic representation: use a higher-order or otherwise physically motivated fit, but validate it against the calculated data.
- Avoid choosing a more complicated fit merely because it has a high R-squared; the key test is whether it preserves the transition energies over the k range relevant to χ^2 .

6. Updated workflow for the k-space $\chi^{(2)}$ calculation

5. Run a k-resolved electronic-structure calculation for the coupled quantum well.
6. Track the relevant electron and heavy-hole subbands consistently across $k_{\parallel}$.
7. Fit the electron subbands to parabolas and extract effective masses if the fit is good over the integration range.
8. Represent the hole subbands with a validated nonparabolic relation or numerical interpolation.
9. At each $k_{\parallel}$, construct the required transition energies $\Delta E_{nm}(k)$.
10. Insert those transition energies into the Eq. 2 resonance denominators.
11. Use the matrix-element treatment chosen in the previous derivation, such as zone-center matrix elements if that approximation has been validated.
12. Apply the radial k-space weight and integrate to the chosen $k_{\max}$.
13. Check convergence and compare against a more direct k-resolved calculation where possible.

$$\chi^{(2)} \sim \int \frac{M(0)}{D_1(k_{\parallel})D_2(k_{\parallel})} 2\pi k_{\parallel} dk_{\parallel}$$

New model decision: The main new piece from this slide is not the integral itself. It is the decision to model $E_e(k)$ and $E_h(k)$ differently because the two classes of subbands do not have the same degree of parabolicity.

7. How this connects to your Nextnano workflow

Needed quantity	What you need from the solver/post-processing	How this slide changes the plan
$E_{e1}(k), E_{e2}(k)$	Electron subband dispersion	A parabolic fit may be sufficient.
$E_{hh1}(k), E_{hh2}(k)$	Heavy-hole subband dispersion	Do not force one constant-mass parabola if it does not fit.
Wavefunctions / matrix elements	State wavefunctions, overlaps, z matrix elements	Previous slide proposed evaluating these at $k = 0$ if their k-dependence is negligible.
$\Delta E_{nm}(k)$	Electron minus heavy-hole transition energy	Uses the fitted electron relation plus the more accurate hole relation.
χ^2	Numerical k integral	Sensitive to both the energy model and k-space weighting.

Validation target: If you are using an 8-band k.p Nextnano calculation, the safest validation is to compare the simple reconstruction against the actual k-resolved energies over the exact k range used in the χ^2 integral.

8. Why holes are more complicated - what the slide does and does not prove

What the slide directly supports: the displayed hole energies are not adequately parabolic in the free direction over the plotted range.

What the slide does not by itself prove: the specific microscopic cause of the nonparabolicity or the unique best fitting function.

Keep source vs. interpretation separate: A common physical reason in multiband semiconductor models is valence-band mixing, but that explanation is additional semiconductor-physics context, not something established by this slide alone.

The implementation question to ask next

"For the hole states, do we want to integrate using the raw/interpolated $k.p$ energies, or do we want to fit them to a specific nonparabolic analytic function?"

That question follows directly from your boss's conclusion and determines how the Eq. 2 code should represent $E_{hh,m}(k_{\text{parallel}})$.

9. Final cheat sheet

Question	Answer to remember
What is being tested?	Whether a constant-effective-mass parabola reproduces $E(k_{\text{parallel}})$.
Electrons?	Approximately parabolic in the shown calculation; effective-mass model is reasonable.
Holes?	Nonparabolic in the shown calculation; use a more faithful $E_h(k)$ relation.
Why do we care?	$E_e(k) - E_h(k)$ sets the optical transition energy in the Eq. 2 denominators.
What does a parabolic fit give?	The effective mass from the curvature.
What changes in the integration?	Electron $E(k)$ may be reconstructed analytically; hole $E(k)$ needs a richer numerical/functional representation.

$$E_e(k), E_h(k) \rightarrow \Delta E(k) \rightarrow \text{Eq. 2 denominators} \rightarrow k\text{-integration} \rightarrow \chi^{(2)}$$

Bottom line: Memorize this: electron dispersion is simple enough for an effective-mass approximation; hole dispersion is not. The hole energy therefore needs to stay more faithful to the multiband k.p calculation before you build the transition energies and integrate $\chi^{(2)}$.

10. Relation to the sequence of your boss's earlier slides

14. Functional Form: identify bands, confined states, and k_{parallel} .
15. Adjusting the Form: convert the k sum into an explicit radial k integral and write denominators in energy form.
16. Energy Relations: propose using zone-center matrix elements and a simple $E(k)$ dispersion.
17. This slide: test the dispersion assumption and discover that electrons and holes need different treatments.

That progression is the key storyline: first define what must be summed, then make the integration explicit, then choose a tractable model for each k -dependent quantity, and finally test whether those approximations are actually valid.